To Forge or Not to Forge: Predicting Task-Level Fine-Tuning Gains from Ten-Example Pilots

Anonymous Author(s)
Affiliation
Address
`email`

Abstract

Fine-tuning a small specialist model is the default path to cheap task-specific capability, yet the upstream decision — is this task worth fine-tuning at all? — is still made by intuition or by paying for the full training run. We ask whether ten labeled examples can answer it in advance. Under a single frozen protocol (Qwen2.5-1.5B-Instruct, fixed LoRA recipe and budgets, greedy decoding, execution-based verifiers), we measure 61 tasks — 11 anchored in the small-model post-training literature and 50 drawn by a pre-registered filter from Super-NaturalInstructions — pairing a ten-example pilot with a full fine-tune whose measured gain serves as ground truth. All analyses were pre-registered. We find: (1) fine-tuning helps far more often than not — 53 of 61 tasks (87%) show positive full-training gains, making "forge" the empirical default in this regime; (2) pilot signals significantly rank gain magnitude (Spearman 0.755, 95% CI [0.60, 0.86]), enabling budget allocation by ranking; (3) the pre-registered binary go/no-go endpoint is not established (leave-one-task-out AUC 0.755, bootstrap CI [0.50, 0.93]), a power limit rooted in the empirical scarcity of no-go tasks itself; and (4) description-only features predict below chance (AUC 0.31) — the decision-relevant information exists only in actually running the pilot. We characterize the no-go tail, including negative transfer on heavily post-trained bases (GSM8K -0.23 despite in-domain training), and release the benchmark, protocol, and all sealed artifacts. These sealed measurements and pre-registered analyses were produced by an autonomous discover-pilot-decide-train-validate system; without that system's contribution the 61-task ground-truth table would not exist.

Keywords: fine-tuning, evaluation, small language models

1 Autonomous research result

Part 1 stub (CFP $\leq$4 pages). Setup, results, and compact framing to follow.

2 System Design and Meta-Analysis

Part 2 stub (CFP $\leq$4 pages; human-authored). Harness, interventions, and verification to follow.

3 Broader Impact

Part 3 stub (CFP $\leq$1 page). Reflections to follow.

A Reproducibility Statement

Stub.

B Disclosure Statement

Agent(s) / model(s) used

- ☐ A human author has curated this work and verified every claim, figure, methodology detail, and reference.
- ☐ A human author takes full accountability for the submission as a human-authored, human-verified artifact.

C References